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Key Points:
- A new constitutive relationship for seismicity rate that includes interactions is derived
- Interacting seismic sources can be modeled as equivalent noninteracting sources
- Applications to fault slip models, aftershock diffusion and secondary triggering are demonstrated

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Constitutive Law for Earthquake Production Based on Rate-and-State Friction: Theory and Application of Interacting Sources

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Abstract The constitutive relationship of Dieterich (1994, https://doi.org/10.1029/93JB02581) for earthquake production has been applied widely to earthquake triggering. The Dieterich theory achieves mathematical simplicity and thus lends itself to practical applications through a number of simplifying assumptions. Perhaps the most suspect and criticized assumption is that seismic sources do not interact. Here I provide an extension of the constitutive framework, using the modified formulation of Heimisson and Segall (2018, https://doi.org/10.1029/2018JB015656) which accounts for source interactions. Moreover, I explore ways to account for earthquake magnitudes and highlight potential directions for future research. I identify conditions under which the constitutive relationship with interactions reduces to the noninteracting Dieterich theory, revealing that an interacting population may be modeled as an equivalent noninteracting population. This result may partly explain the success of the Dieterich theory in spite of assuming no interactions. These special conditions reveal that if the magnitude distribution (e.g., the b value) is altered by a stress perturbation, then the Dieterich model may either underpredict or overpredict the seismicity rate. Furthermore, the interaction model grants insight into the physical interpretation of the background seismicity rate, which is frequently considered to be a declustered rate. However, the results suggest that interactions do not change the total cumulative number of events over a long time span ($t \gg t_a$). Finally, I apply the interaction theory to fault slip models, aftershock diffusion, and secondary triggering as illustrative examples.

1. Introduction

It is well known that earthquakes cluster in space and time. The clustering may be due to a common locally or regionally applied stress perturbation, shared seismogenic structures, or direct source-to-source interactions. Determining how these different factors integrate and manifest in observable seismic activity is far from trivial, and this uncertainty may significantly alter how we interpret and understand observed seismic activity. Thus, furthering physical models of earthquake triggering and clustering is of great importance to natural hazard monitoring that frequently relies heavily on detecting and locating microseismic activity.

The seminal work of Dieterich (1994) (which will be referred to here as Dieterich's model or theory) has provided substantial insight into earthquake clustering, for example, by providing a physical basis for Omori's law of aftershocks. This work was revisited by Heimisson and Segall (2018) and rederived in a modified form (this will be referred to as modified Dieterich's model or theory if such distinction is needed). Heimisson and Segall (2018) furthermore explored many assumptions of the model and generally concluded that they are usually reasonable. However, the assumption of noninteracting sources was not investigated in detail.

Here I address the problem of interactions in a more general manner than has been done before. Previously, Ziv and Rubin (2003) investigated a special case of source interactions. For a coplanar distribution of sources where the net contribution of each source to the effective stressing rate is approximately constant and the magnitude distribution is not altered by the stress perturbation, they showed that the Dieterich model applies in spite of interactions. I verify their theoretical findings, shown in detail later.

This study is a direct continuation of Heimisson and Segall (2018), but it is useful to review some of the principal findings. Heimisson and Segall (2018) presented a modified Dieterich theory where they attained

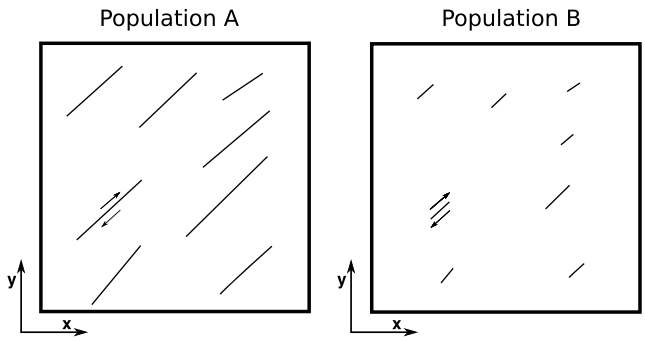

Figure 1. Two isolated populations that are identical except that in population A the sources interact since source size is large compared to the distance between sources, and in B they do not interact because the source size is small compared to the distance between the sources.

integral expressions for both the cumulative number of events N and seismicity rate R

$$\frac{N}{r} = t_a \log\left(\frac{1}{t_a}\int_0^t K(t')\mathrm{d}t' + 1\right), \tag{1}$$

$$\frac{R}{r} = \frac{K(t)}{1 + \frac{1}{t_a}\int_0^t K(t')dt'}, \tag{2}$$

where $t_a = A\sigma_0/\dot{\tau}_r$ is the characteristic decay time of aftershock activity that was identified by Dieterich (1994). A is a constitutive parameter that relates to the rate dependence of friction, and σ_0 is the initial normal stress acting on a population. $K(t)$ is a stress history-dependent integral kernel function that can be written out explicitly in terms of shear $\tau(t)$ and normal stress $\sigma(t)$

$$K(t) = \exp\left(\frac{\tau(t)}{A\sigma(t)} - \frac{\tau_0}{A\sigma_0}\right)\left(\frac{\sigma(t)}{\sigma_0}\right)^{\alpha/A} \simeq \exp\left(\frac{S(t)}{A\sigma_0}\right). \tag{3}$$

The approximation in equation (3) is the *Coulomb stress approximation*, which was discussed in detail by Heimisson and Segall (2018), where $S(t) = \Delta\tau(t) - (\tau_0/\sigma_0 - \alpha)\Delta\sigma(t)$ and $\Delta\tau(t)$ and $\Delta\sigma(t)$ are the time-dependent changes in stress (such that $\tau(t) = \tau_0 + \Delta\tau(t)$). Equations (1) and (2) are exactly equivalent to the theory of Dieterich (1994) in the limit when changes in normal stress are small.

It is useful to start by reviewing the concept of a population of seismic sources and the underlying assumptions. The concept of a population (also called a distribution) is the building block of the Dieterich theory, but it is rarely explicitly explained in the literature and may thus confuse a reader that is unfamiliar with this type of modeling. A population of seismic sources is in this work defined as a collection of faults or fault patches that have not failed seismically and that occupy a volume of the crust. This volume cannot be selected arbitrarily—it must be small enough such that the applied stress field and the fault orientations are approximately constant. Furthermore, the initial shear and normal stress (τ_0 and σ_0, respectively) and frictional parameter A (related to the direct effect) should be spatially constant. Under constant background stressing rate $\dot{s}_r$, it is usually assumed that the sources fail at constant rate r; although as shown by Heimisson and Segall (2018) the model can be generalized for nonconstant background seismicity rate. All these conditions are generally not met unless the volume is much smaller than the characteristic wavelengths of the spatial variability of the stresses and relevant parameters. This spatial scale may very well be smaller than the spatial scale of each source. It can thus be useful to consider the Dieterich model to be valid at a point or in an infinitesimal volume. In this case, r becomes the background seismicity rate per unit volume and does not represent a physical seismicity rate of a population of seismic sources and must be understood probabilistically as the expected seismicity rate per unit volume at a specific spatial coordinate $\mathbf{x}^p$. The seismicity rate ($R(t, \mathbf{x}^p)$) or cumulative number of events ($N(t, \mathbf{x}^p)$) can be integrated in space accounting for heterogeneity. This is often done (Hainzl et al., 2010), for example, to model the effect of spatially heterogenous stresses (Helmstetter & Shaw, 2006).

It is broadly true that any physical theory that accounts for interactions of particles, fields, or defects is more complicated than its noninteracting counterpart. Let us consider two identical populations, subject to the same spatially uniform stresses, same value of A and same background seismicity rate at the same background stressing rate but with one important difference: In population A the sources interact, but in population B the sources do not interact (Figure 1). This means that the Dieterich theory is valid for population B but perhaps not for A. We have assumed that the two population have the same background seismicity rate at constant background stressing rate. However, does that mean that their seismicity rate is the same for an arbitrary stressing history? If that is true, we may model the behavior of population A as if it was population B. Thus, we model the system as an imaginary noninteracting population, which is vastly simpler than accounting for all the interactions. In this paper, I show that indeed this is true, under certain assumptions, and I use that to construct a model where interaction *within* each population need not be computed and only the interaction *between* different populations are computed. This will frequently be referred to as the *interaction model or theory* in this paper.

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This paper has three main parts: First, I investigate interactions within a single population and identify the conditions where the population can be modeled as an equivalent noninteracting population. Second, I use that result to construct a model that accounts for interaction between populations. Third, I show applications of that model.

2. Theory

Heimisson and Segall (2018) showed that we may relate time to instability t of the Nth event for arbitrary Coulomb stressing history $S_a(t)$ to the time to instability under background conditions defined by the background Coulomb stressing history $S_b(t)$ using the following equation:

$$\int_0^t \exp\left(\frac{S_a(t')}{A\sigma_0}\right) \mathrm{d}t' = \int_0^{\mathcal{B}(N)} \exp\left(\frac{S_b(t')}{A\sigma_0}\right) \mathrm{d}t'. \tag{4}$$

$\mathcal{B}(N)$ is a function that describes the time to instability of the Nth source. Here I will for simplicity assume a constant background rate r under background conditions such that $\mathcal{B}(N) = N/r$. Note that r is not a declustered rate but can be thought of as a long-term average (section 4.4). Note that it is straightforward to include a different $\mathcal{B}(N)$ function.

2.1. Interaction Within a Uniform Isolated Population

Let us investigate a volume of the crust that contains of population of seismic sources that satisfies all conditions previously stated to define a population. Furthermore, we assume that there are no seismogenic sources outside this population; thus, it may be considered isolated. The Coulomb stress change acting on the population after N sources have nucleated is

$$S(t) = S'(t) + \sum_{i=1}^{N} \Delta S(i), \tag{5}$$

where $S'(t)$ is remotely *applied* Coulomb stress change, which is approximately constant over the spatial scale of the volume containing the population and $\sum_i \Delta S(i)$ is the change in Coulomb stress change caused by each failed source. For the sake of simplicity, I shall assume that each source contributes a spatially constant Coulomb stress change on all other sources. We may thus regard $\Delta S(i)$ as in some sense the average stress imposed by the ith on its neighbors that together collectively form the population that we want to describe. In Appendix A I show that this assumption of spatially uniform interacting stress is not necessary.

In the case of arbitrary stressing history in equation (4) the applied stress component can be written as

$$S'_a(t) = \Delta\tau(t) - \mu\Delta\sigma(t) \text{where} \, \mu = (\tau_0/\sigma_0 - \alpha); \tag{6}$$

however, for the background stressing, the population is subject to constant shear and/or normal stressing rates: $\dot{\tau}_r$ and $\dot{\sigma}_r$, respectively,

$$S'_b(t) = \dot{s}_r t = (\dot{\tau}_r - \mu\dot{\sigma}_r)t. \tag{7}$$

In order to make comparison with the findings of Ziv and Rubin (2003), I further simplify equation (5) by setting the stress magnitude as a constant $\Delta S(i) = \Delta S$. A continuous approximation to the sum gives

$$S(t) = S'(t) + \Delta SN. \tag{8}$$

The treatment of interactions in equation (8) is equivalent to what was done in the analytical model proposed by Ziv and Rubin (2003). Equation (8) under background conditions reduces to $S_b(t) = \dot{s}_r t + \Delta Srt$; thus, the interactions effectively change the background stressing rate from $\dot{s}_r$ to $\dot{s}_r + \Delta Sr$. Hereinafter I will drop the subscripts $_a$ and $_b$ and generally represent the stressing histories as the sum of the applied stress and interaction stress. For example, $S'(t)$ will represent the arbitrary applied stressing history and $\dot{s}_r t$ is the applied background stress.

Equating the background condition integral to the arbitrary stressing integral, as in equation (4), and assuming that the sources fail at constant rate under background condition gives

$$\int_0^t \exp\left(\frac{S'(t')}{A\sigma_0} + \frac{\Delta SN(t')}{A\sigma_0}\right) \mathrm{d}t' = \int_0^{N/r} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0} + \frac{\Delta St'r}{A\sigma_0}\right) \mathrm{d}t'. \tag{9}$$

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Differentiating both sides with respect to t gives

$$\exp\left(\frac{S'(t)}{A\sigma_0} + \cancel{\frac{\Delta SN}{A\sigma_0}}\right) = \exp\left(\frac{\dot{s}_r N/r}{A\sigma_0} + \cancel{\frac{\Delta SN}{A\sigma_0}}\right)\frac{1}{r}\frac{\mathrm{d}N}{\mathrm{d}t}. \tag{10}$$

The interaction term cancels from both sides. It is easy to see that we are left with a separable ODE, which has the solution shown in equation (1) for the approximate kernel function in equation (3). This means in the case of an isolated population where all fields are uniform then the population can be modeled as an equivalent noninteracting population with the same background seismicity rate under the same background stressing rate. Insight into this counterintuitive result can be gained from equation (9), where on the right-hand side the interactions alter the effective stressing rate under background conditions and on the left-hand side the interactions add (or subtract if $\Delta S < 0$) to the stressing history of each source. Thus, if interactions are included, they effectively increase the stress (assuming $\Delta S(i) > 0$), which promotes higher rates of seismicity. However, the interaction will then also increase the effective background stressing rate, which decreases the seismicity more quickly because the characteristic decay time t_a is decreased. Remarkably, in this simple case, the two cancel completely. If $\Delta S(i) < 0$, then the stress is reduced, but the effective stressing rate is also reduced such that again the two effects cancel. It is important to note that this does not mean that interactions on small distance scales cannot result in temporal clustering of events. This result simply means that if the sources fail at constant rate under background conditions, or more generally at times $\mathcal{B}(N)$ (see Appendix A), in two populations, one with internal interaction and one without internal interaction, then they will produce the same seismicity rate when subject to the same arbitrary stressing history. This result is contingent on several conditions that are reviewed and explicitly stated at the end of this section.

2.1.1. Stress-Dependent Earthquake Interactions

One of the important assumptions is that perturbing the stress does not alter the magnitude distribution of events. In other words, those events will have the same magnitude when they occur under the constant stressing rate conditions as they do under the arbitrary stressing history. If this assumption does not hold, then the cancelation shown above will not occur and the Dieterich model does not accurately describe a population with interactions. To illustrate this point, let us consider a very simple model for how the moment might systematically change.

Let us imagine two exact duplicates of the same fault: let one copy produce an earthquake under constant background stressing rate and another nucleate under a perturbed stress state. A number of factors could alter the difference in moment release. However, it may be reasonable to assume that for small differences in the perturbing stress and background stress some sort of a linearized version of that relationship holds (whatever that may be). Let us assume that the stress transfer from one source to another is linearly proportional to the moment then we may propose a modified version of equation (5)

$$S(t) = S'(t) + \sum_{i=1}^{N}[\Delta S(i) + c\cdot(S(t_i) - S(i/r))], \tag{11}$$

where $\Delta S(i)$ is the average contribution to the stress field of the i-th source to all the other sources under background stressing conditions and c is a constant that relates the difference in Coulomb stress change at the time of failure in the perturbed and background case for the Nth source to the change in interaction stress.

To explore the influence of changing the moment, I simplify (11) further where I only consider the applied stress field to alter the moment and approximate the sum as a continuous integral

$$S(t) = S'(t) + \int_0^N \Delta S(N') + c\cdot(S'(t(N')) - \dot{s}_r(N'/r))\mathrm{d}N'. \tag{12}$$

For the Omori case of instantaneous stress step $\Delta S'$ at $t = 0$ and then constant stressing rate $\dot{s}_r$ one finds

$$\frac{R}{r} = \exp\left(\frac{\Delta S' + \dot{s}_r t - \dot{s}_r N/r}{A\sigma_0}\right)\exp\left(c\frac{\Delta S'N - \dot{s}_r N^2/2r + \dot{s}_r\int_0^N t(N')\mathrm{d}N'}{A\sigma_0}\right), \tag{13}$$

which can be solved numerically. At time $t = 0$ this model is identical to the Dieterich model, which is expected as interactions have not yet occurred. Early on, the sources fail at higher stress than they would

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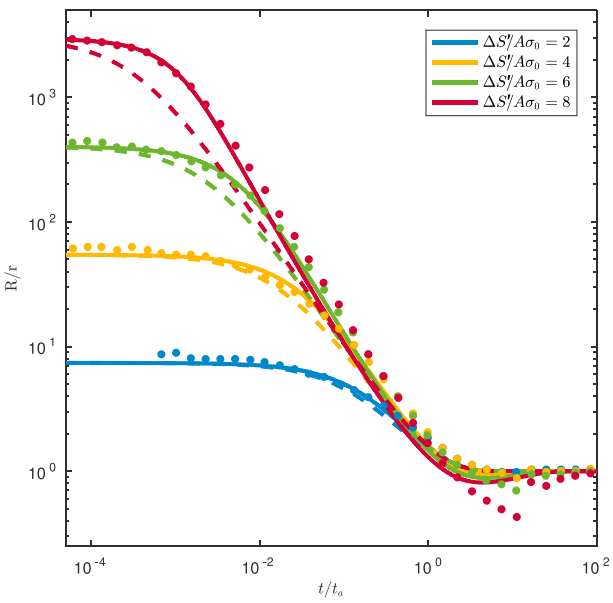

Figure 2. Seismicity rate followed by change in stess $\Delta S'$ at $t = 0$ and constant background stressing rate $\dot{s}_r$. Comparison of equation (13) (solid line) with the Dieterich's theory (dashed line), which is also equivalent to an interacting model without change in source moment. Dots show results of numerical simulations of Ziv and Rubin (2003; Figure 6: top left panel) which account for both interactions and source healing. Equation (13) similar to the simulations of Ziv and Rubin (2003) predicts extended period of near-constant seismicity rate initially (compared to noninteracting theory). Furthermore, both models predict seismicity rate $R < r$ for times $t \sim t_a$. The value of $c = 33.3 \cdot 10^{-6}$ was found by fitting with the results of Ziv and Rubin (2003) and provides excellent agreement for $t \ll t_a$.

under background conditions. The simple model I have proposed suggests that their moment release is higher, increasing the seismicity rate initially and delaying the characteristic 1/time drop in seismicity rate. Later on in the sequence near $t \sim t_a$ the sources fail at lower stress than they otherwise would have, which results in decreased moment and seismicity rate. This highly simplified model shows surprising agreement with the results of Ziv and Rubin (2003); see Figure 2. This suggests that the changes in moment reported by Ziv and Rubin (2003) may be partly explained by the simple model proposed in equation (12); however, it is unclear how much of the discrepancy can be explained by an incomplete description of the interaction or the lack of accounting for state healing, which may be important near $t \sim t_a$ (Heimisson & Segall, 2018).

As was previously mentioned, I have assumed here for simplicity and clarity that the stress perturbation from each source to be constant within a certain volume; however, these are not realistic assumptions. In a more general case each source applies different magnitudes of stress change to its neighbors and the stress change is spatially variable through the population of the sources. This case is analyzed in Appendix A, where the same cancelation occurs assuming that the source moment is unchanged by the perturbing stresses. This means that within this population the model reduces to Dieterich's model. In Appendix A, I show that this cancelation also occurs when time to instability is an arbitrary function of the source number: $t = \mathcal{B}(N)$, as in equation (4).

It is appropriate to end this section by summarizing the conditions for the cancelation of the interaction terms in Appendix A or equation 10.

1. The stress field, background stressing rate, and A are constant in space, and initial stresses are the same for all sources in the population. These conditions are part of the definition of a *population of seismic sources*.
2. Coulomb stress approximation is valid.
3. The magnitude statistics are not altered by the stress perturbation.

2.2. Interacting Populations

In the previous section, I analyzed a highly idealized case where I postulated that there is an isolated volume containing a population of seismogenic sources. In that case, there is no effect from source interaction on the rate (under certain conditions) and the population can be modeled as an equivalent population of noninteracting sources that have the same background seismicity rate under the same background stressing conditions. Now I explore a more general case where each population interacts with its neighbors. This analysis is based on the following assumptions:

1. There exists a length scale which defines a volume of the crust, containing a *population of seismogenic sources*, where the applied stress field, the initial stresses for each source, A, and the background stressing rate are all approximately constant and do not deviate significantly from some average value (as in section 2.1).
2. The characteristic dimension of the sources is generally small compared to the characteristic length scale of the volume containing the population. Thus, the triggering effect of large events are not well approximated as point sources may not be accounted for accurately.
3. The stress change caused by the failure of the ith source in population q will cause approximately constant average stress change in a spatially separated population p. The stress interaction is noted as $\Delta S^{pq}(i)$. Thus, the first superscript index indicates in which population the stress change is felt and the second the population where the source failed.

In section 4.2 and Appendix B I discuss how interactions can be formulated where these assumptions are not strictly necessary. In particular, the third point implies that if population volumes are systematically decreased, then due to spatial variability in the gradient of the interaction stresses, the overall rate of earth-

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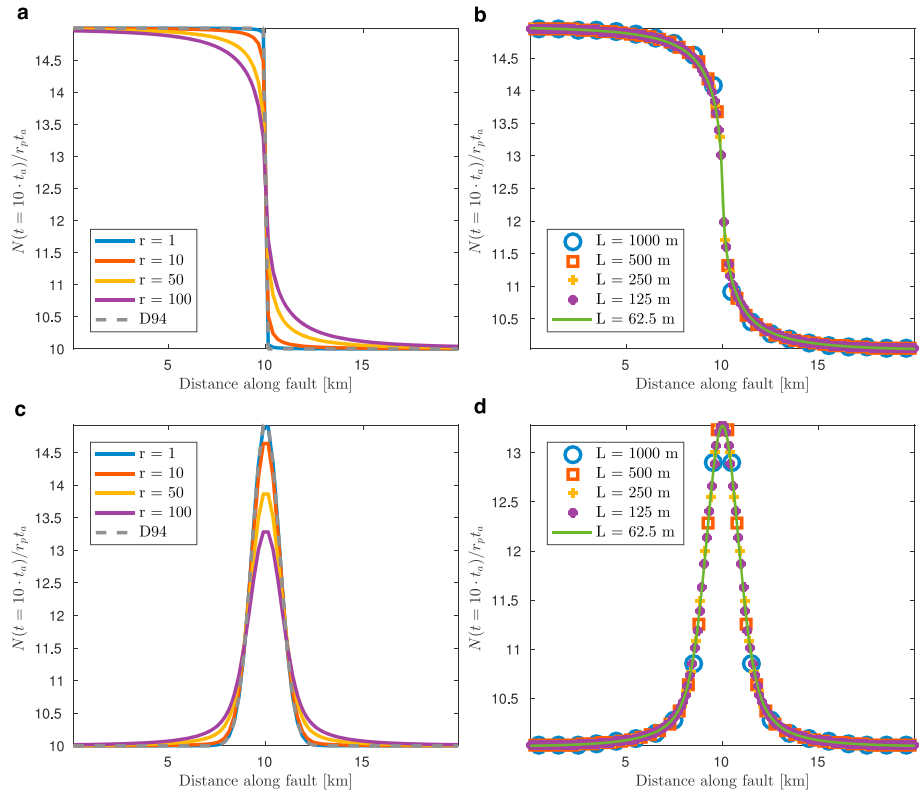

Figure 3. Cumulative number of events after the first 10 km of a fault have been perturbed by a uniform stress step of 0.5 MPa (a and b) and a Gaussian shaped perturbation with peak value of 0.5 MPa (c and d). Here $t_a = 10$ years, $A\sigma_0 = 0.1$ MPa, and the units of the background rate r are in events per year for the entire fault (20 km), which is divided evenly between populations. The interaction stress is computed from equation (25). (a, c) The cumulative numbers of events at each population, at time $t = 10 \cdot t_a$ with $L = 250$ m (length of each population; see Appendix B), as a function of distance along the fault. N is nondimensionalized by $r_p t_a$, where r_p is the background rate in each population. The influence of interactions increases at larger r with increasing deviations from the Dieterich model. (b, d)The same as (a) and (c), respectively, except with fixed $r = 100$ events per year, but with varying L. Symbols mark the center of each population and show convergence as L becomes smaller.

quakes may not converge. However, scaling arguments in Appendix B and simulations in Figure 3 show that there is convergence if the seismic moment released scales with the volume of the population.

For implementation, we assume that $\Delta S^{pq}(i)$ can be calculated from a point source located at the center of population q. Thus, effectively assuming that all the sources that fail in q will add up coherently and be well represented as a single point source at the center of the population with a cumulative moment release depends on N^q (the cumulative number of events in population q). Assuming a point source makes stress change linearly dependent on the seismic moment which is convenient for computation (e.g., Okada, 1992).

The stress change in population p is thus given by

$$S^p(t) = S^{p\prime}(t) + \sum_{q=1}^{M} \int_0^{N^q} \Delta S^{pq}(N')\mathrm{d}N', \tag{14}$$

where M is the total number of populations. The constitutive relationship for population p can now be derived

$$\int_0^t \exp\left(\frac{S^{p\prime}(t')}{A^p\sigma_0^p} + \frac{\sum_{q=1}^{M}\int_0^{N^q(t')} \Delta S^{pq}(N')\mathrm{d}N'}{A^p\sigma_0^p}\right)\mathrm{d}t' = \int_0^{N^p/r^p} \exp\left(\frac{\dot{s}_r^p t'}{A^p\sigma_0^p} + \frac{\sum_{q=1}^{M}\int_0^{t'r^q} \Delta S^{pq}(N')dN'}{A^p\sigma_0^p}\right)\mathrm{d}t'. \tag{15}$$

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Differentiating both sides gives the rate in population p

$$\frac{R^p}{r^p} = \frac{1}{r^p}\frac{\mathrm{d}N^p}{\mathrm{d}t} = \exp\left(\frac{S^{p\prime}(t) - \dot{s}_r^p N^p/r^p}{A^p\sigma_0^p}\right)\exp\left(\frac{1}{A^p\sigma_0^p}\sum_{q=1}^{M}\int_{\frac{N^p r^q}{r^p}}^{N^q}\Delta S^{pq}(N')\mathrm{d}N'\right). \tag{16}$$

Equation (16) defines a system of M coupled nonlinear ordinary differential equations describing the rate of earthquakes in each population. For computational implementation it is important to note that the sign of the integral is positive if $N^q > \frac{N^p r^q}{r^p}$ and negative if $N^q < \frac{N^p r^q}{r^p}$ assuming that $\Delta S^{pq} > 0$. If for populations p and q the scaled cumulative number of events is the same, that is $N^p/r^p = N^q/r^q$, then those populations are no longer coupled and could be regarded as one population with a background rate of $r^q + r^p$ assuming that the two are spatially connected. The condition that the scaled cumulative number of events is the same $N^p/r^p = N^q/r^q$, apart from background stressing conditions when this is always true by definition, would typically require that the stressing history in the two populations is the same, and the following fields τ_0, $A\sigma_0$, and $\dot{s}_r$ are the same and spatially constant. The model thus shows that it is consistent with the cancelation effects that have been shown to occur within each population. If all the required fields are constant within a certain volume, it can be split into subvolumes and subpopulations without it affecting the model showing the consistency of the model. For example, if all the aforementioned conditions are met in all populations, then all interaction terms cancel and the seismicity rate for each population could be modeled using the Dieterich model. Alternatively, they could all be modeled as one population with a background rate $\sum_i r^i$.

Due to coupled and highly nonlinear nature of equation (16), solving the system of ODEs analytically for a physically interesting case is very difficult. However, to gain insight into how this model differs from Dieterich's model, I shall look at the case where only population p is perturbed by stress change $\Delta S'$ at $t = 0$. Furthermore, I assume that t is small such that population p has not significantly interacted with any other populations; therefore, $N^q = r^q t$, except when $q = p$. Finally, I assume that ΔS^{pq} is constant with respect to N^q. Under these assumption, one can show that the cumulative number of events in population p is

$$\frac{N^p}{r^p} = \frac{A^p\sigma_0^p}{\dot{s}_r^{eff}}\log\left(e^{\frac{\Delta S}{A^p\sigma_0^p}}\left(e^{\frac{\dot{s}_r^{eff}t}{A^p\sigma_0^p}} - 1\right) + 1\right) \tag{17}$$

and seismicity rate

$$\frac{R^p}{r^p} = \frac{e^{\frac{\Delta S' + \dot{s}_r^{eff}t}{A^p\sigma_0^p}}}{e^{\frac{\Delta S}{A^p\sigma_0^p}}\left(e^{\frac{\dot{s}_r^{eff}t}{A^p\sigma_0^p}} - 1\right) + 1}, \tag{18}$$

which is precisely the same equation that Dieterich's theory predicts, with the exception that the background stressing rate $\dot{s}_r$ has been replaced by an effective stressing rate $\dot{s}_r^{eff}$ given by

$$\dot{s}_r^{eff} = \dot{s}_r^p + \sum_{q=1}^{M}\Delta S^{pq} r^q. \tag{19}$$

The effect of interactions between populations, in this case, is to change the background stressing rate. Note that in the limit $t \to 0$ the instantaneous rate of earthquakes is the same for the two models; this is to be expected since if no event has nucleated, then no interactions can occur. The characteristic decay time is $t_a = \dot{s}_r^{eff}/A^p\sigma_0^p$, which will be shorter if the sum in equation (19) is positive or longer if it is negative. This example shows that considering interactions can resolve a paradox that arises in the Dieterich theory (modified or unmodified) as the background stressing rate goes to zero. From equation (2) it can be seen that if $\dot{\tau}_r \to 0$, then the seismicity rate will not decay with time. Clearly, the model does not behave in a physically realistic manner in the limit of the background stressing rate going to zero. However, interactions contribute to the effective stressing rate such that even though the tectonic stressing rate could be near zero the interactions will contribute to the decay of the seismicity rate to background levels. For example, in intraplate settings, where the tectonic stressing rate is probably extremely small, one might argue that the Dieterich model is not appropriate. Our analysis, however, suggests that the interaction model presented here could be applied in those cases. The interaction model may, therefore, have particular relevance to problems related to injection-induced seismicity that can occur far from plate boundaries (Ellsworth, 2013). See further discussion in section 4.3.

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2.2.1. Interactions That Are Discrete in Time

In equation (16) the sum of stress transfers from sources in population q as experienced by the pth population was approximated as a continuous integral

$$\sum_{i=1}^{N^q} \Delta S^{pq}(i) \approx \int_0^{N^q} \Delta S^{pq}(N')\mathrm{d}N'. \tag{20}$$

This approximation is typically appropriate when computing the cumulative number of events given some applied stressing history by solving the system described in equation (16). However, in many applications of the Dieterich theory the cumulative number of events (or seismicity rate) is not simulated. Instead, observed seismicity rate and cumulative number of events are used to extract the changes in stress (e.g., Dieterich et al., 2000; Helmstetter & Shaw, 2009; Inbal et al., 2017; Ziv, 2012).

The approximation in equation (20) is not necessary mathematically but convenient when N is allowed change continuously with time. If directly applying observed seismicity to extract the stress history, then treating interactions as time discrete events may be more attractive. Following a similar approach as in section 2.2, one finds

$$\frac{R^p}{r^p} = \exp\left(\frac{S^{p\prime}(t') - \dot{s}_r^p N^p/r^p}{A^p\sigma_0^p}\right)\exp\left[\frac{1}{A^p\sigma_0^p}\sum_{q=1}^{M}\left(\sum_{i=1}^{N^q}\Delta S^{pq}(i) - \sum_{i=1}^{\lfloor \frac{N^p}{r^p} r^q \rfloor}\Delta S^{pq}(i)\right)\right], \tag{21}$$

where $\lfloor\rfloor$ represents the floor function (rounds a number down to nearest integer). If $N^q = 0$ or $\lfloor \frac{N^p}{r^p} r^q \rfloor = 0$, then the corresponding sum is defined to be zero. From equation (21) it is easy to write the applied stress change at population p , $S^{p\prime}(t)$, as a function of the seismicity rate at p and the cumulative number of events at all populations

$$S^{p\prime}(t) = A^p\sigma_0^p \log\left(\frac{R^p}{r^p}\right) - \sum_{q=1}^{M}\left(\sum_{i=1}^{N^q}\Delta S^{pq}(i) - \sum_{i=1}^{\lfloor \frac{N^p}{r^p} r^q \rfloor}\Delta S^{pq}(i)\right) + \frac{\dot{s}_r^p}{r^p}N^p. \tag{22}$$

Equation (22) is comparable to equation 55 in Helmstetter and Shaw (2009), where the rate at $R(t = 0)$ is assumed to be the background rate. However, here source interactions have been taken into account. In application of the Dieterich theory for joint inversion the role of earthquake interactions and secondary events is often cited (e.g., Inbal et al., 2017; Segall et al., 2013) as an element of uncertainty in determining the applied stress field. Equation (22) provides a possible resolution as it allows us to separate the externally applied stress field $S^{p\prime}(t)$ and the changes that occur through source interactions. In section 3.4 I show a practical example of how this theory can be used.

3. Application

3.1. Modeling Earthquake Clustering

To model earthquake clustering where interactions are considered, it is important to account for the magnitude distribution of events. This could be done in a few ways, for example, by sampling magnitudes from the Gutenberg-Richter (GR) distribution before starting the simulation. This means that the Nth source in the pth population has a seismic moment that has been determined from stochastic sampling before the earthquake rates or cumulative number of events are simulated. This method would render different results for each sampling of seismic moments. Below I explore a deterministic method of including the earthquake magnitude distribution, which will not capture the influence of large triggered events as in a stochastic sampling but can give insight into the overall effect of stress transfer if all interactions are assumed to be caused by events with an average seismic moment release of the distribution.

3.1.1. Inclusion of the GR Distribution

If it is assumed that a population has a maximum and minimum seismic moment, then we attain a probability distribution for the seismic moment in the form of a power law Pareto distribution (Kagan, 2002).

$$\phi(M) = \frac{M_{\max}^{\beta} M_{\min}^{\beta}}{M_{\max}^{\beta} - M_{\min}^{\beta}}\beta M^{-1-\beta} \text{ where } M_{\min} \le M \le M_{\max}, \tag{23}$$

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where $\beta = 2b/3$. For $\beta < 1$ the Pareto distribution has no mean value, which is why I have introduced a truncated distribution. Under the assumptions of a maximum and minimum moment we can compute a mean moment for the distribution (Kagan, 2002)

$$\langle M \rangle = \frac{\beta}{1-\beta} \frac{M_{\min}^{\beta} M_{\max} - M_{\min} M_{\max}^{\beta}}{M_{\max}^{\beta} - M_{\min}^{\beta}}. \tag{24}$$

For one given event in population q then the stress change in all other populations is

$$\Delta S^{pq} = f(\mathbf{x}^p, \mathbf{x}^q) \langle M \rangle, \tag{25}$$

which can be used in equation (16), where $f(\mathbf{x}^p, \mathbf{x}^q)$ is the influence of a point source of unit moment in population q acting in population p. In this study where I use equation 25 I will assume $b = 1$, a maximum magnitude of 6, and minimum magnitude of 1.

3.2. Triggering on a 1-D Fault

To demonstrate the influence of interactions in the model that has been described in previous sections with the average seismic moment model in equation (24) I set up a simple example simulation for a 1-D fault. Initially, $f(\mathbf{x}^p, \mathbf{x}^q)$ in equation (25) is computed assuming a source of unit length along strike but is much longer in up and down dip directions, thus approximating a line source. The influence of this source is evaluated at centers of other populations (Figure B1). Then $f(\mathbf{x}^p, \mathbf{x}^q)$ represents the influence of a source at population p evaluated at population q for a unit slip and unit shear modulus divided by the area of the source, which here is just equal to its downdip length since it is of unit length in strike direction.

To highlight the effects of the interactions, I solve the system of equations given by (16) for spatially and temporally simple stressing. At time $t = 0$ the first 10 km of a 20-km-long fault is subject to stress step of $\Delta S' = 0.5$ MPa. However, the only applied stress to the other half of the fault is a constant background tectonic stressing rate. This is qualitatively similar to the scenario explored in simulations by Ziv (2006a). Since the Dieterich model contains no interactions the seismicity rate would only change in the first 10 km, but in reality one would expect that due to interaction the last 10 km would also see change in seismicity rate. This is indeed what is seen in the simulations shown in Figure 3a, where due to stress transfer the seismicity rate and therefore cumulative number of events in the last 20 km of the fault is increased. Perhaps more surprisingly the seismicity rate decays faster in time in the perturbed region near the interface of the perturbed and nonperturbed parts of the fault. This is due to higher effective background stressing rate (compared to the Dieterich model) that is not fully compensated by increased stress transfer. Further away from the interface, the conditions for an isolated population at spatially constant stress (outlined in section 2.1) are approximately met and the interaction model behaves like the Dieterich model. The results in Figure 3 show how interactions may explain delayed aftershocks in regions where coseismic stress changes are small (Ziv, 2006b).

The interaction model, unlike the Dieterich model, has a nonlinear dependence on the background seismicity rate r (Figure 3). This is simple to understand since the higher the background rate of seismicity, the more earthquakes are triggered and thus the greater is the effect of interactions. The same holds true for any parameter that controls the cumulative number of events. It follows that the instantaneous seismicity rate at time $t = 0$ will always be the same in the interaction model as it is in the Dieterich model because if no events have nucleated, then no interactions have occurred.

Figures 3b and 3d establish convergence of simulations with spatial refinement of L, the characteristic length scale of the volume of the crust that hosts each population. Note, however, that Appendix B explains that under certain conditions such convergence may not be expected and the length scale L must satisfy restrictions as are mentioned in the beginning of section 2.2.

In the case of the narrow Gaussian stress perturbation (Figure 3c) the populations near the peak stress will produce fewer events than the same populations in the Dieterich model. This is the same effect as seen in Figure 3a near the change in applied stress and can be understood by examining equation (18). Populations at peak stress are exposed to higher effective background stressing rate than their counterparts in the Dieterich model and thus produce fewer events. In simplistic terms, the populations at peak applied stress, transfer more interaction stress to their neighbors than they receive. They cannot, therefore, compensate for

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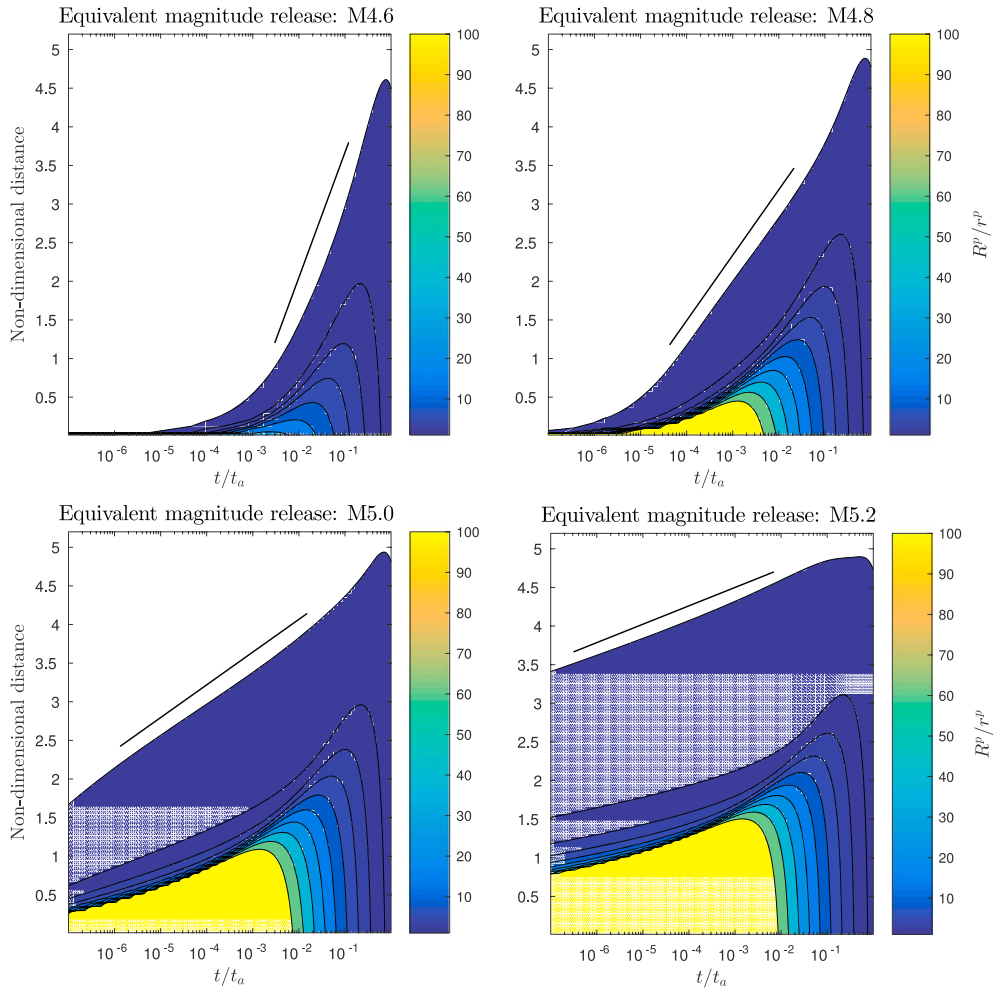

Figure 4. Scaled seismicity rate outside an area perturbed at time zero. Only stress transfer in this part is due to secondary triggering, and thus, the original Dieterich model would not predict any changes in the seismicity rate. In all cases the front of increased seismicity rate shows a linear trend on a log time scale (black line) over a several orders of magnitude in time, thus suggesting that such triggering may be observed as log(t) migration of seismicity. The vertical distance axis is nondimensionalized be a characteristic rupture radius $R = (7M_0/(16\Delta\tau_c))^{(1/3)}$ (Lay & Wallace, 1995), where M_0 is the total moment released from the perturbed population and the stress drop $\Delta\tau_c$ is taken as 5 MPa.

the higher effective background stress rate and produce fewer earthquakes than the Dieterich model predicts. The opposite is true for populations at low stress, which produce more events than predicted by the Dieterich theory. Curiously, the total number of events appears to be conserved. For example, if the cumulative number of events in all populations is summed in Figure 3a or Figure 3c, then the Dieterich model and the interaction model will produce the same number of events at times much larger than t_a. This implies conservation of the overall number of events between the two models, although they can be distributed differently. This suggests that the Dieterich model should be able to predict reasonably well the final cumulative number of events on a regional scale, although it may not fully capture the final distribution of events locally or transient changes on shorter timescales. This result has implications for the physical interpretation of the background seismicity rate (see section 4.4).

3.3. Aftershock Diffusion

In the previous section (e.g., Figure 3), the seismicity rate is elevated in areas that are not perturbed by stress changes as a consequence of secondary stress transfer. This migration of seismicity qualitatively resembles diffusion. It has been proposed for a long time that aftershock areas grow with time in some places (e.g., Tajima & Kanamori, 1985). This is often referred to as *aftershock diffusion*. Studies that have attempted to empirically quantify the characteristics of aftershock diffusion have proposed $l = Dt^H$, where l is a characterstic distance between mainshock and aftershocks, t is time, and H is a power law exponent that characterizes

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the growth rate of the aftershock zone. D is a constant that likely depends on magnitude, tectonic setting, and more. Generally H is found to be small where Marsan et al. (2000) estimated $H \sim 0.1$–0.3. However, Helmstetter et al. (2003) found even weaker growth with $H \sim 0$–0.1 and a similar value was estimated by Huc and Main (2003) with $H \sim 0.06$. Peng and Zhao (2009) investigated the early time migration of aftershocks and determined that the 2004 Parkfield earthquake aftershocks migrated with $l \sim \log(t)$, which was suggested to be caused by afterslip. It is worth noting that there may not be a discrepancy between a power law migration $l \sim t^H$, where H is small, and a log time migration $l \sim \log(t)$, since $(t^H - 1)/H \rightarrow \log(t)$ as $H \rightarrow 0$, and therefore, a $\log(t)$ migration can be similar to power law migration if H is sufficiently small.

It was recognized by Dieterich (1994) that due to stress decay with distance from a main shock his model could predict similar behavior. However, Marsan et al. (2000) concluded that the Dieterich model would generally predict $H \gtrsim 0.3$ and would thus not explain the very small exponents that have been estimated. Triggering by cascading earthquakes has been proposed as a possible mechanism of such diffusion and has been analyzed in the context of the epidemic-type aftershock sequence model (Helmstetter & Sornette, 2002). Clearly, if that is the dominant mechanism of the aftershock diffusion, it will not be captured by the Dieterich model since it assumes that sources do not interact. The model presented in this study should, however, account for this effect.

Figure 4 shows increased number of events that are caused by secondary stressing with time. Similarly as in Figure 3a I perturb a part of a 1-D fault and leave the other parts unperturbed. The perturbed populations (assumed to be the first 5 km of a 40-km-long fault) thus generate an excess moment release. However, I impose no externally applied stress gradients; this is clearly unphysical but is done to isolate the effect of the cascading style of triggering. The moment release is altered by changing the magnitude of the stress step at time zero.

Figure 4 demonstrates that the growth of the aftershock zone has a clear linear trend on a log time scale, thus suggesting that the aftershock region due to cascading triggering of events may show $l \sim \log(t)$ trend in data, such as observed by Peng and Zhao (2009), and such migration may not necessarily be driven by afterslip.

3.4. Extracting Stress Change From Seismic Data

The Dieterich seismicity rate theory has been implemented in joint inversions of geodetic data and triggered seismicity to attain slip distributions from earthquakes (e.g., Ziv, 2012) or aseismic transients (e.g., Inbal et al., 2017), where equations derived by Helmstetter and Shaw (2009) or Dieterich et al. (2000) are used to solve for the stress changes that result from slip on a fault. Then stress changes can be related to slip through elastic Green's functions (e.g., Okada, 1992). However, in order to attain the applied stress to the system, we need to distinguish between stress changes from interactions, which may occur on secondary faults, and the applied stress changes on the main fault. If the two are separated there should be a direct correspondence between slip on a major fault as detected from triggered seismicity and geodesy.

Equation (22) allows us to solve for the applied stress ($S^{p_r}(t)$) acting on a population, if the interaction stresses ($\sum_{q=1}^{M} \sum_{i=1} \Delta S^{pq}(i)$) are computed based on earthquake locations and estimated seismic moment, thus separating applied and interaction stresses.

In order to test the differences between the Dieterich model and equation (22), I perform a slip inversion for the $M6$ 2004 Parkfield earthquake (e.g., Johanson et al., 2006; Liu et al., 2006). This earthquake is well documented in an area with a good record of seismicity both before and after the earthquake; furthermore, the event was relatively simple since it was mostly local to a single strand of the San Andreas Fault system. I will first only use earthquakes, without geodetic data, to illustrate differences that arise from including and disregarding interactions. Then I perform a joint inversion of GPS and stress changes derived from seismicity while keeping everything in the inversion scheme the same. This reveals how including earthquakes may resolve some issues that arise from nonuniqueness in inversions that only used geodetic data.

From the NCEDC (2016) catalog, I retrieve all earthquakes from 1970 to 2016 in the vicinity of the Parkfield earthquake. I remove events smaller than $M1.5$ to minimize the effect of long-term changes in network sensitivity. All earthquakes within 5 km from the fault plane are binned into rectangular areas that are 1 km in height (down dip) and 1 km in length (along strike). I take all events in 5-km distance from the fault; however, a vast majority of events are located within $\sim$ 1-km distance from the fault plane, which is of the same order as the patch dimensions. The earthquake sources in each population can thus be expected to be subject to approximately constant stress change. Furthermore, I set $A = 0.001$ and σ_0 is taken as the difference

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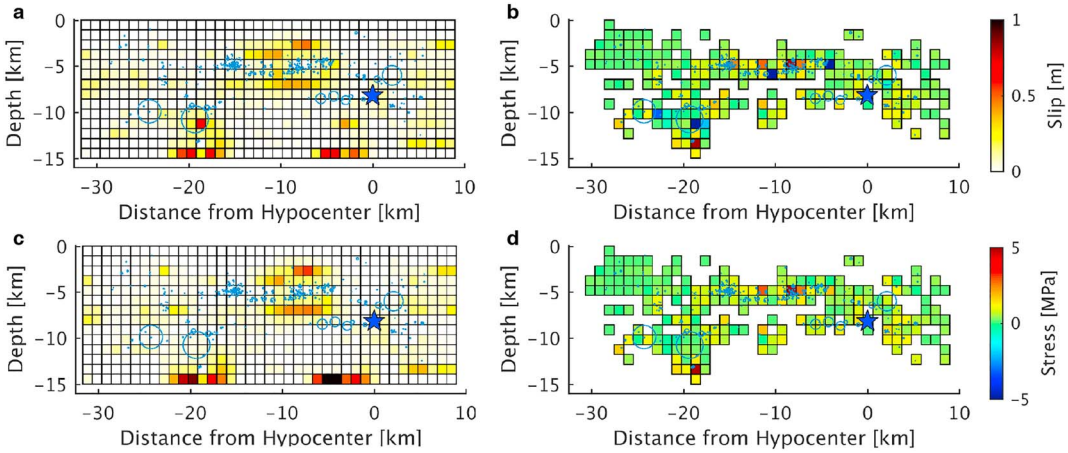

Figure 5. Slip distribution and stresses 3 days after the main shock derived from aftershock data. (a) Slip inversion of stresses in (b) that is attained through equation (22). (c) Slip inversion of stresses in (d) where interactions are ignored (equivalent to Dieterich, 1994). Both slip models are subject to the same smoothing constraints and are constrained to have the same total seismic moment, which is equivalent to a *M*6 earthquake. In both cases the data and independent model parameters are exactly the same. Models produce a variance reduction >95% compared to the null model where the interaction model fits slightly better (difference about 3%). Blue dots and circles show earthquakes of *M*1.5 or larger within 3 days from the main shock. Circle size represents the size of event if they were circular cracks with a stress drop of 5 MPa. Large blue star shows the location of the main shock hypocenter.

in lithostatic pressure and hydrostatic pore pressure. The seismicity from 1970 until the Parkfield events in 2004 is used to establish a background seismicity rate in each bin. Each bin (rectangular patch in Figure 5) is furthermore assumed to represent a dislocation patch on the fault. Thus, observed seismicity is used to infer the applied stress changes using equation (22) then stress changes are related to slip via a matrix of influence coefficients computed for the dislocation grid. Bins or patches that do not have any measurable background seismicity rate or no seismicity after the Parkfield earthquake are assumed to contain no data; however, the dislocation representing that bin is still included in the grid, which makes the problem underdetermined. To regularize the slip inversion, I apply a Laplacian smoothing constraints on each patch and constrain the total seismic moment to be equal to that of the Parkfield earthquake, corresponding to *M*6. The weight of the smoothing coefficient is chosen such that the model predicts 99% reduction in variance of the observationally derived predicted stress changes. To compute interactions, I assume purely strike-slip sources and compute the influence of a rectangular dislocation with a unit area (effectively a point source) and unit seismic moment using Green's functions derived by Nikkhoo et al. (2016). Their solutions do not have artifacts along dislocation lines, which were found to be problematic with other Green's function solutions.

Comparison of the slip model with and without interactions (Figure 5) shows that the two models are in general agreement, but there are patches where the models differ significantly. The most notable difference is that the interaction model predicts larger changes in the applied stress and predicts negative stress changes (stress drops) in some locations. In the stress derived from the Dieterich theory virtually no stress drop is predicted. This is because measuring seismicity rates that are lower than the background rate is very difficult. However, in the interaction model, the stress drops come about from the computed interaction stress, which in some places are large, but may not result in a corresponding increase in seismicity. This means that the applied stress must be negative to begin with. The location of slip and slip amplitude is in broad agreement with the coseismic slip model derived using different methods such as Johanson et al. (2006), Liu et al. (2006), and Ziv (2012).

It is worth noting that a few aftershocks, in particular, a *M*5 event located at (−20, −11; Figure 5) have source dimensions that are larger than the patch size, assuming a circular crack and the assumed stress drop gives realistic source dimensions. Clearly, if this is the case, then the conditions in Appendix B are violated. However, it leads to a rather interesting prediction in the slip model (Figure 5a). If the interaction model and the Dieterich model are compared, we see that the interaction model predicts significant slip at the source region of the *M*5 aftershock, whereas the Dieterich model places almost no slip in the same area. This is a

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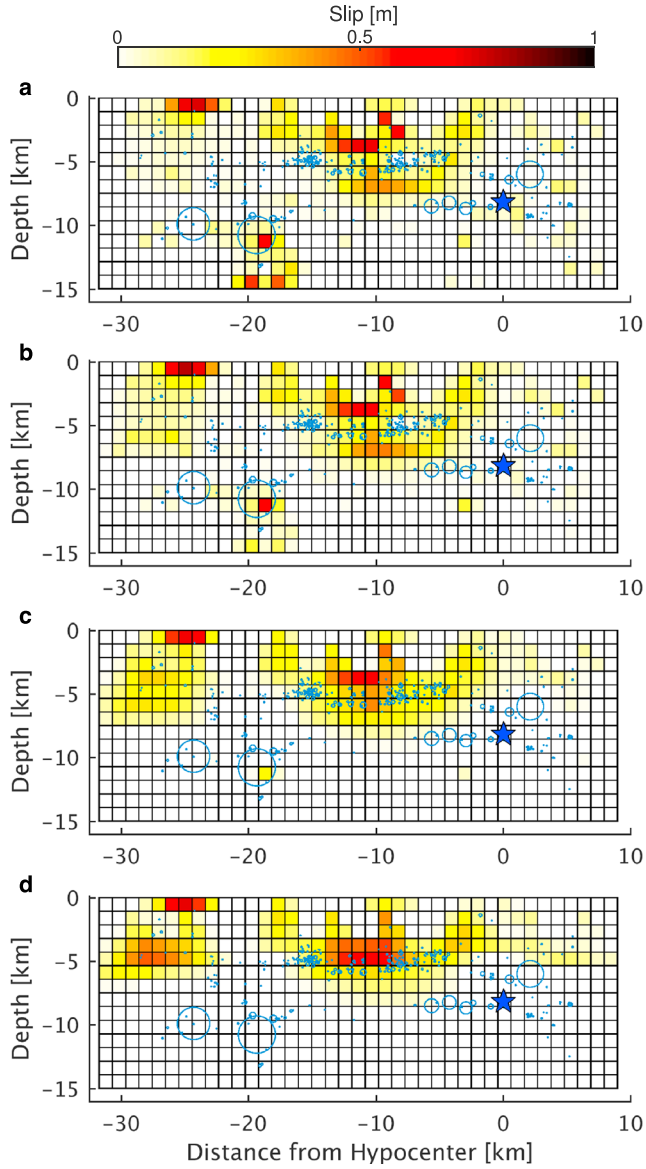

Figure 6. Slip distribution from joint inversion of GPS and earthquakes where interactions are included. (a) Preferred model that well predicts both GPS data and stress changes inferred from earthquakes. (b–d) Weight of stress change reduced by factors of 0.5, 0.1, and 0.01, respectively, compared with the preferred model in (a). All slip models produce fit to the GPS data with about 95% reduction in variance, but only (a) and (b) can acceptably explain the stress distribution inferred from earthquakes. Model (d), which is mostly constrained by GPS, produces significantly worse fit to the stress changes then the null model.

consequence of the interaction model predicting that the applied stress in this region must be negative since the *M*5 event (approximated as a point source) did not trigger a significant change in seismicity.

To demonstrate the usefulness of including stress changes inferred from aftershocks, I perform a joint inversion of coseismic GPS displacements from Houlié et al. (2014) and stress changed inferred in Figure 5b. I apply the same Laplacian smoothing as was set in Figure 5, but the weight of the stress changes is systematically varied until a preferred model is reached where both GPS and stress changes are fitted with over 95% reduction in variance. Then I explore systematically decreasing the weight of the stress changes in the joint inversion compared to the preferred model which reveals systematic changes in the inferred slip model (Figure 6). Although the fit to the GPS data in all cases is comparable, the agreement with stress changed derived from aftershocks is highly variable. This strongly suggests that including aftershock data in fault slip models, as has been outlined here, may greatly reduce the uncertainty that arises from nonuniqueness

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in geodetically derived fault slip models and conditions the slip model to be consistent with the aftershock distribution in a quantitative manner.

4. Discussion

4.1. Stress-Dependent Changes in *b* Value

One of the more surprising results in this work is the link between the validity of the Dieterich model and shifts in the magnitude distribution of events as a consequence of a stress perturbation. This implies that if there is a detected change in b value from background conditions, we may find that the Dieterich model either overpredicts or underpredicts the seismicity rate. Increase in earthquake magnitudes corresponds to higher seismicity rate and decrease in magnitudes to lower seismicity rate. The interaction stresses will be proportional to the change in seismic moment (difference in seismic moment at background conditions compared to perturbed conditions). Thus, small shifts in magnitudes may not be consequential. In agreement with Figure 2, Ziv and Rubin (2003) showed in simulations that after a stress step the magnitudes will first increase, but later on, they decrease and are lower than background levels, eventually returning to background values. Nevertheless, they found in most cases a good agreement with the Dieterich theory, which is expected if the magnitude distribution is not altered (and other conditions discussed in, e.g., section 2.1 are met).

In this study, I make the assumption that the magnitude distribution is not changed by the stress transfer, which allows us to derive the interaction model (section 2.2). The impact of this assumption needs further investigation. To fully account for this effect, a constitutive relationship relating stress change to shifts in the magnitudes of events is needed.

4.2. Relaxing Assumptions in the Interaction Model

The interaction model derived in section 2.2 is dependent on several assumptions that may not always be reasonable. As was discussed before, we require that the stress perturbation does not change the source magnitudes. The length scale of the populations must satisfy conditions in Appendix B unless the problem can be formulated such that the seismic moment released will scale with the volume of the population. Addressing the problem of when the perturbation shifts magnitudes is not simple, because we currently do not know how the seismic moment release might be changed by perturbing the stress. Second, accounting for interactions at close distances is difficult due to the strong dependence on how sources are located with respect to each other. If someone were to derive a constitutive relationship between stress history and change in seismic moment release, the second issue of computing close range interactions could be solved simply by reducing the population length scale such that it is comparable to the length scale of a seismic source. Doing so means considering the constitutive relationship for earthquake production (e.g., equation 16) as being valid at a point, which was discussed in section 1. At small enough volumes we do not need to think about interactions within each volume since the volume will generally not contain more than one source and perhaps only a partial source. However, as has been mentioned before, a convergence with decreasing population volume size is only attained if the seismic moment scales with the volume.

The method for including the GR distribution in section 3.1.1 allows the seismic moment released to scale with volume size, which can be seen from the convergence in Figure 3. However, the approach taken in section 3.4, where the observed earthquake magnitudes are binned into predefined volumes, is subject to restrictions on length scales. Stochastically sampling the seismic moment of each source in each population would also be subject to these limitations. Thus, to remove the length scale dependence in these cases, a method is needed to compute a seismic moment release density from data, which considers that one event may not be exclusively located within one volume. That research direction is beyond the scope of this study.

4.3. Limit of Low Background Stressing Rate

If we suppose that the background stressing rate is zero ($\dot{s}_r = 0$), then the dependence of seismicity rate on stress in a noninteracting theory is simply $R = rK(t)$ (equation 2). Thus, if a stress perturbation is applied to the population and that perturbation does not decay with time, such as for an aftershock sequence, then the seismicity rate is permanently and exponentially elevated. Clearly, this is a nonphysical example since a constant rate of seismicity could not be maintained for a long period of time without some background stressing prior to the perturbation of the stress field. However, this end-member case may be worth considering in intraplate settings, where the background stressing rate is extremely small. The dominant control over the decay time of aftershocks in such regions could be the effective stressing rate that has been shown

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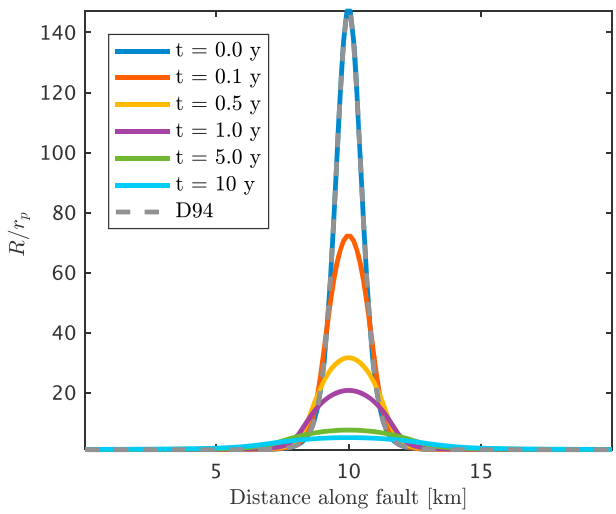

Figure 7. Scaled seismicity rate per population following a Gaussian stress perturbation with peak stress 0.5 MPa (Similar to Figures 3c and 3d). In this simulation $A\sigma_0 = 0.1$ MPa and the background rate is $r = 100$ events per year, which is divided evenly for all population on the fault. The interaction stress is computed from equation (25). Unlike previous simulation $\dot{s}_r = 0$, and thus, $t_a = \infty$. The Dieterich model (both modified and original) does not change with time, but the interaction model decays due to interactions creating an effective background stressing rate.

to arise from interactions as in equation (19). Furthermore, for long sustained aftershock sequences, aftershock interaction will become more important and migration of aftershocks away from the epicenter with time may occur, which could be similar to diffusion, although probably with an exponent of less than 1/2. To illustrate this point, I have done a simulation for a Gaussian perturbation, where the background stressing rate $\dot{s}_r = 0$ (Figure 7). The Dieterich model, either modified or original, does not decay with time, whereas the interaction model spreads out and decays in a qualitatively similar manner to a diffusion process. Indeed, long sustained aftershock sequences have been observed for intraplate earthquakes that have shown space-time characteristics similar to a diffusion process (Mandal et al., 2016).

4.4. Physical Interpretation of *r*

Simulations in this study show that the net cumulative number of events over a long time ($t \gg t_a$) is the same with or without interactions. This has also been true for 2-D and 3-D simulations that I have done but are not shown here. Often in applications of the Dieterich model, the observed seismicity is declustered, based on the argument that the Dieterich theory does not account for the secondary triggering. The findings in this paper argue against this interpretation, where secondary triggering appears not to alter the cumulative number of events but effectively redistributes stresses and alters the spatial and temporal distribution of events. The background rate in the Dieterich model should thus be taken as the long-term average with interactions, not as a declustered rate. Different declustering algorithms may, furthermore, render variable results and introduce model bias since they are based on different criteria and can generate variable results when used on the same data set (Marsan & Lengline, 2008). In most models of earthquake clustering (such as the epidemic-type aftershock sequence model), there is the general idea of binary triggering; either a source is triggered or it is not triggered. This implies that if a source fails from secondary triggering in an aftershock sequence, it was not influenced by the original mainshock. However, it is expected that the mainshock also altered the time to instability of the secondary source. Declustering neglects this effect and will remove earthquakes that may have been very close to instability even without the secondary interaction. The theory in this paper accounts for the total stressing history, both applied stresses and interactions on each source.

It is worth emphasizing that these results do not mean that short-range interactions cannot result in transient temporal and spatial clustering. Under the assumption of constant background seismicity rate under background conditions such clustering will be smoothed out in time by definition. Choosing the background conditions to produce constant seismicity rate with interactions may seem unrealistic. Nevertheless, it represents a steady state where energy is dissipated at the same rate as it is introduced into the system by the background stressing. This is not a stable steady state, and the system can demonstrate organization and critical behavior; nevertheless, it is probably the best and simplest approximation we can make to define the initial conditions of the modeled populations. I note that clustering can be artificially introduced by adding temporal correlation into the background time to instability ($t = \mathcal{B}(N)$); however, that would not get at the physics of such clustering. It may be a fruitful avenue of future research to consider how the model can be modified to reproduce clustering at background conditions in a nonartificial manner.

5. Conclusions

In this study, I addressed what may be the most criticized limitation of the Dieterich theory, which is the assumption of noninteracting sources. Intuitively, we understand that interactions must give rise to secondary stressing that may, for example, cause earthquakes to migrate. Previous work has indicated that such triggering through cascading of events is important (Marsan & Lengline, 2008). I have presented a new physical model that accounts for interactions; the primary result is equation 16, which may be regarded as an extension of the Dieterich theory where all populations couple through stress interactions.

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Among the most surprising findings is that conditions exist such that a population of sources may be modeled as an equivalent noninteracting population. This finding may explain why the Dieterich theory appears to be in good agreement with observations in spite of lacking interactions. Furthermore, these conditions define a spatial discretization such that interaction within a population need not be considered and only interaction between populations are important. This applies under the conditions listed at the end of section 2.1.

I have demonstrated applications of the new interaction theory, which suggests that interactions may somewhat influence fault slip models that are constrained from stress changes derived from aftershocks. Furthermore, the results show that including such stress changes in a geodetic fault slip inversion is a good method of quantitatively constraining a slip model that is consistent with an aftershock distribution. The new interaction model has been applied to problems of seismicity migration due to secondary triggering. According to simulations, such migration has qualitative characteristics of a diffusion process and a log-time spatial growth of the region of increased seismicity rate. Finally, I challenge the view that declustering of data should be done when applying the Dieterich model. Declustering implies that secondary events are independent of the total stressing history, and the findings of this study suggest that the long-term cumulative number of events is independent of interactions on a regional scale.

Appendix A: Nonconstant Stress Interactions Within a Population

We start with a population of sources that interact. The volume containing the sources is small enough such that externally applied stress change $S^{'}(t)$ is constant, as well as the initial shear stress τ_0 and normal stress σ_0, background stressing rate $\dot{s}_r$, and parameter A. The stress change acting on the Nth source to fail once $N - 1$ sources have failed is

$$S_N(t) = S'(t) + \sum_{q=1}^{N} \Delta S_{Nq}, \tag{A1}$$

where $\Delta S_{NN} = 0$ since a source cannot interact with itself. Note that I use subscript ΔS_{pq} when indicating interactions between sources within a population but superscript ΔS^{pq} to indicate interactions between different populations. Now I allow noninteger values of N so that I can regard time to instability as a continous variable that is equivalent to time. It can be convenient within this time continuous description to represent the stressing history of the Nth source as a smoothly varying function, although it should be noted that this is not necessary. I approximate the sum by an integral over function $\Delta S_N(q)$ such that the cumulative stress at the Nth source at time t is

$$S(t) = S'(t) + \int_0^{N(t)} \Delta S_N(N')\mathrm{d}N', \tag{A2}$$

where $N^{'} = N(t)$ when $t = t_{\mathrm{inst}}(N)$, that is, when instability of the N source is reached. Interaction stresses change continuously with source number $N^{'}$ in $\Delta S_N(N^{'})$; thus, equation (A2) can be thought of as accounting for the more general case when not all the slip occurs instantaneously if $\Delta S_N(N^{'})$ is chosen to account for aseismic slip leading up to instability. Similar to equation (9), I find

$$\int_0^t \exp\left(\frac{S'(t')}{A\sigma_0} + \frac{\int_0^{N(t')} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right)\mathrm{d}t' = \int_0^{N/r} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0} + \frac{\int_0^{t'r} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right)\mathrm{d}t'. \tag{A3}$$

Differentiating both sides with respect to t

$$\exp\left(\frac{S'(t)}{A\sigma_0} + \frac{\int_0^{N(t)} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right) = \exp\left(\frac{\dot{s}_r N/r}{A\sigma_0} + \frac{\int_0^{N(t)} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right)\frac{1}{r}\frac{\mathrm{d}N}{\mathrm{d}t}, \tag{A4}$$

from which we see a clear cancelation of the part of the stress that corresponds to source-to-source interaction. Below I show this again for the purely discrete case, which may be more intuitive to many readers.

Interestingly, this result is not specifically a consequence of assuming a constant background rate of seismicity under constant stressing rate. We may hypothesize any relationship between the time to instability

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and source number under constant stressing rate $t = \mathcal{B}(N)$ in which case equations A3 and A4 become, respectively,

$$\int_0^t \exp\left(\frac{S'(t')}{A\sigma_0} + \frac{\int_0^{N(t')} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right)\mathrm{d}t' = \int_0^{\mathcal{B}(N)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0} + \frac{\int_0^{\mathcal{B}^{-1}(t')} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right)\mathrm{d}t', \tag{A5}$$

$$\exp\left(\frac{S'(t)}{A\sigma_0} + \frac{\int_0^{N(t)} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right) = \exp\left(\frac{\dot{s}_r \mathcal{B}(N)}{A\sigma_0} + \frac{\int_0^{N(t)} \Delta S_N(N')\mathrm{d}N'}{A\sigma_0}\right)\frac{\mathrm{d}\mathcal{B}(N)}{\mathrm{d}t}. \tag{A6}$$

To better understand this remarkable cancelation, let us also derive this result using a different approach, which may be more intuitive and less abstract. Consider two populations A and B. A is in every way the same as B except that the sources interact in A but not in B (Figure 1). This could be due to the distance between sources that is large compared to the source size in B, but in A the two length scales are comparable.

Before the first source fails ($N = 1$) the populations are in every way identical. The failure condition for the first source in A and B is (following equation 14 in Heimisson & Segall, 2018)

$$\left(\frac{A}{H\dot{\delta}_0}\right)_1^A = \left(\frac{A}{H\dot{\delta}_0}\right)_1^B = \int_0^{(1/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t' = \int_0^{t_1^A = t_1^B} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t'; \tag{A7}$$

thus, the failure condition of the two sources is the same, and therefore, the time of instability of the first source in both population is the same, that is, $t_1^A = t_1^B$. Now once the first source has failed, these failure conditions are different for the second source. For population B we have

$$\left(\frac{A}{H\dot{\delta}_0}\right)_2^B = \int_0^{(1/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t' + \int_{(1/r)}^{(2/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t', \tag{A8}$$

whereas for population A

$$\left(\frac{A}{H\dot{\delta}_0}\right)_2^A = \int_0^{(1/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t' + \exp\left(\frac{\Delta S_{21}}{A\sigma_0}\right)\int_{(1/r)}^{(2/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t', \tag{A9}$$

where for simplicity I take the stress transfer from source 1 to 2, that is, ΔS_{21}, to occur instantaneously, which is likely to be a good approximation in nature. The time to instability for the second source ($N = 2$) in populations B and A under general stressing history $S'(t)$ is

$$\begin{aligned}
&\int_0^{t_1^A} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t' + e^{\left(\frac{\Delta S_{21}}{A\sigma_0}\right)}\int_{t_1^A}^{t_2^A} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t' \\
&= \int_0^{(1/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t' + e^{\left(\frac{\Delta S_{21}}{A\sigma_0}\right)}\int_{(1/r)}^{(2/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t' \\
&\int_0^{t_1^B} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t' + \int_{t_1^B}^{t_2^B} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t' \\
&= \int_0^{(1/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t' + \int_{(1/r)}^{(2/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t'.
\end{aligned} \tag{A10}$$

where t_1^B and t_2^B are the times of instability for source 1 and source 2, respectively, in B and the same notation applies for population A.

Using equation A7 and canceling the stress step term from both sides gives

$$\begin{aligned}
\int_{t_1^A}^{t_2^A} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t' &= \int_{(1/r)}^{(2/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t', \\
\int_{t_1^B}^{t_2^B} \exp\left(\frac{S'(t')}{A\sigma_0}\right)\mathrm{d}t' &= \int_{(1/r)}^{(2/r)} \exp\left(\frac{\dot{s}_r t'}{A\sigma_0}\right)\mathrm{d}t',
\end{aligned} \tag{A11}$$

which requires $t_2^A = t_2^B$ because the factor $\exp(\Delta S_{21}/A\sigma_0)$ occurs on both sides of equation A10 and $t_1^A = t_1^B$. Using equations A7 and A11, one can show in a similar manner that the instability time of the third source

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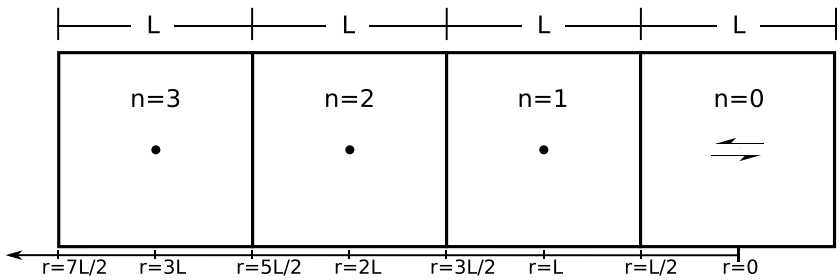

Figure B1. Each box represents a volume of the crust that contains a population of sources and has a characteristic length dimension of L. I assume that volume $n = 0$ has an event that perturbs the stresses in volumes $n = 1, 2, 3$

is the same in both populations $t_3^A = t_3^B$ and then so on for any source number. This leads to the same conclusion as after equations A4 that a system without interactions, but same background seismicity rate under background conditions, is equivalent to an interacting system with the same properties.

Appendix B: Length Scales in the Interaction Model

In the beginning of section 2.2 assumptions were made, which should hold true for the interaction model to be valid. Here I apply scaling arguments to derive a length scale for a volume containing a single population so that these assumptions are met.

First I derive a characteristic length scale L_{cI} for which the interaction stresses in each nth volume due to a source in volume $n = 0$ (Figure B1) can be considered constant. That is, at a certain distance the gradient in stress is small enough such that the perturbation is approximately constant in each volume.

For a circular crack with stress drop of $\Delta\tau_c$ and radius R, the change in stress $\Delta\tau$ over distance $r > R$ is approximately (Dieterich, 1994)

$$\Delta\tau = \Delta\tau_c \left[\left(1 - \frac{R^3}{r^3} \right)^{-1/2} - 1 \right]. \tag{B1}$$

Equation (B1) is only valid in the same plane as the crack itself lies but that is also where the changes in stress are going to be largest so this configuration can be considered an upper limit to the change in stress in the medium surrounding the crack. If $\Delta\tau_c$ is taken as fixed (i.e., a typical value of 5 MPa), then the crack radius can be related to the seismic moment $R^3 = 7M_0/(16\Delta\tau_c)$ (Lay & Wallace, 1995)

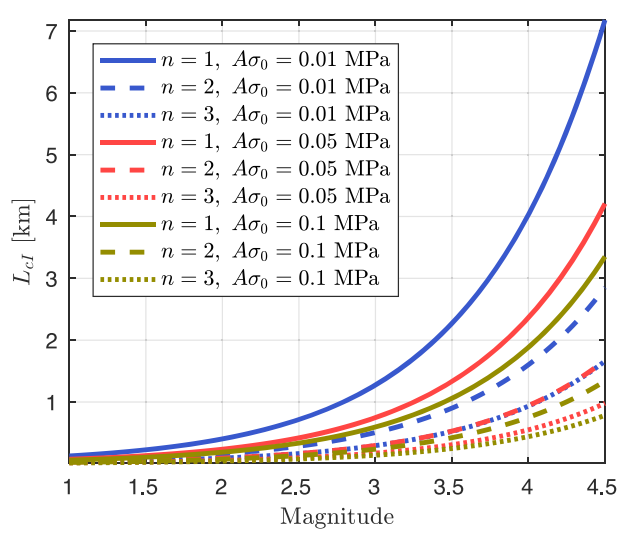

Figure B2. Solutions to equation B2 for different values of n, $A\sigma_0$, and cumulative magnitude of events in volume $n = 0$ in Figure B1. Here the stress drop is taken as $\Delta\tau_c = 5$ MPa, but it should be noted that the solution is not highly sensitive to the value of $\Delta\tau_c$.

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The kernel K (equation 3) suggests that $A\sigma_0$ is a natural measure of significant change in stress. Thus, if the gradient evaluated at the center of each box $\left|\frac{\partial}{\partial r}\Delta\tau(r = nL)\right| \lesssim A\sigma_0/L$, then the stress change from interactions can be regarded as approximately constant. When $L = L_{cI}$, the equality holds, we find

$$\left|\frac{\partial}{\partial r}\Delta\tau(r = nL_{cI})\right| = \frac{21}{32}\frac{M_0}{(nL_{cI})^4\left(1 - \frac{7M_0/(16\Delta\tau_c)}{(nL_{cI})^3}\right)^{3/2}} \lesssim \frac{A\sigma_0}{L_{cI}}. \tag{B2}$$

The characteristic length L_{cI} strongly depends on the value of $A\sigma_0$. Typical values reported when applying the Dieterich theory are in the range from 0.01–0.1 MPa (Hainzl et al., 2010). In Figure B2 I solve equation B2 as function earthquake magnitude in volume $n = 0$ (Figure B1).

In order for the theory in section 2.2 to be accurate, the characteristic length of a volume containing a population L must be chosen such that $L > L_{cI}$. For example, Figure B2 suggests that if $A\sigma_0 = 0.05$ MPa, and the cumulative seismic moment released corresponds to an event of M4.0, then $L \sim 2.3$ km is the lower limit to the characteristic length scale for the volume. However, there is also an upper limit to the length scale of each volume. This is set by the gradient in the stress field $S^{'}(r, t)$. This requirement sets another characteristic length scale L_{cS}, which is natural to define as the distance over which the applied stress field changes by $A\sigma_0$. If this condition is met, then the cancelation shown, for example, in Appendix A occurs. Thus, in order for the theory derived in section 2.2 to be valid, the characteristic length of the volume containing each population L must satisfy

$$L_{cI} < L < L_{cS}. \tag{B3}$$

If such a length scale does not exist, perhaps due to large gradients in stress or very large triggered events, then the modeling approach outlines in section 2.2 will be inaccurate. However, it is likely that the approach is still more accurate than completely ignoring interactions.

Although a point source is probably a less accurate representation of an earthquake source than a circular crack, it is worth investigating that case since point sources are convenient in computing the interactions between volumes. Much simpler scaling arguments than what is derived in equation (B2) where the source is assumed to be a point source suggest the following criteria for the theory being valid

$$\frac{M_0}{(nL)^4} \lesssim \frac{A\sigma_0}{L} \Rightarrow \left(\frac{M_0}{n^4 A\sigma_0}\right)^{1/3} \lesssim L_{cI}, \tag{B4}$$

which gives similar value for L_{cI} as equation (B2). In simulations in section 3.2 the seismic moment released from each population decreases as L increases since the total background seismicity rate for the entire fault is assumed constant. This implies a dependence of the cumulative seismic moment on the length scale. We can write $M_0 = \bar{m}_0 L^3$, where $\bar{m}_0$ is the average seismic moment density within the volume. Substitution of $M_0 = \bar{m}_0 L^3$ in equation (B4) or (B2) reveals that dependence on the length scale cancels, which implies convergence as $L \to 0$. This convergence is also seen in numerical solutions of equation 16 (Figures 3b and 3d).

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